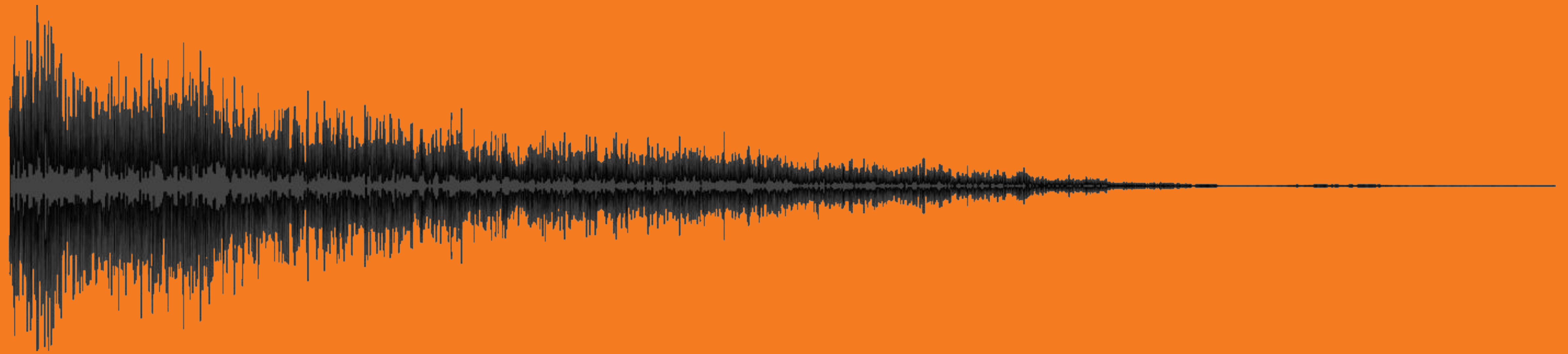
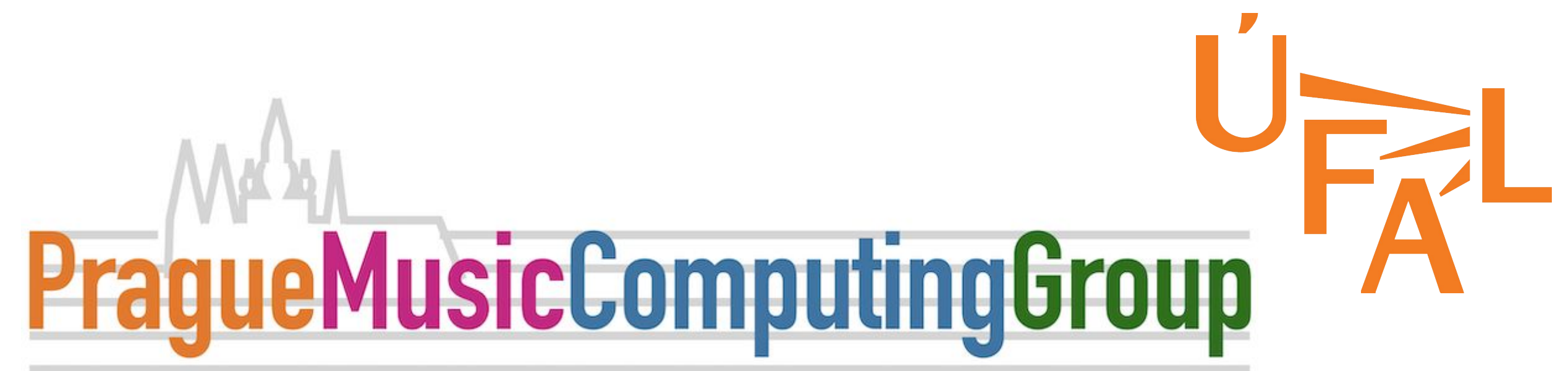




Generating Drum Kit Samples using Machine Learning

Author: Patrik Backo
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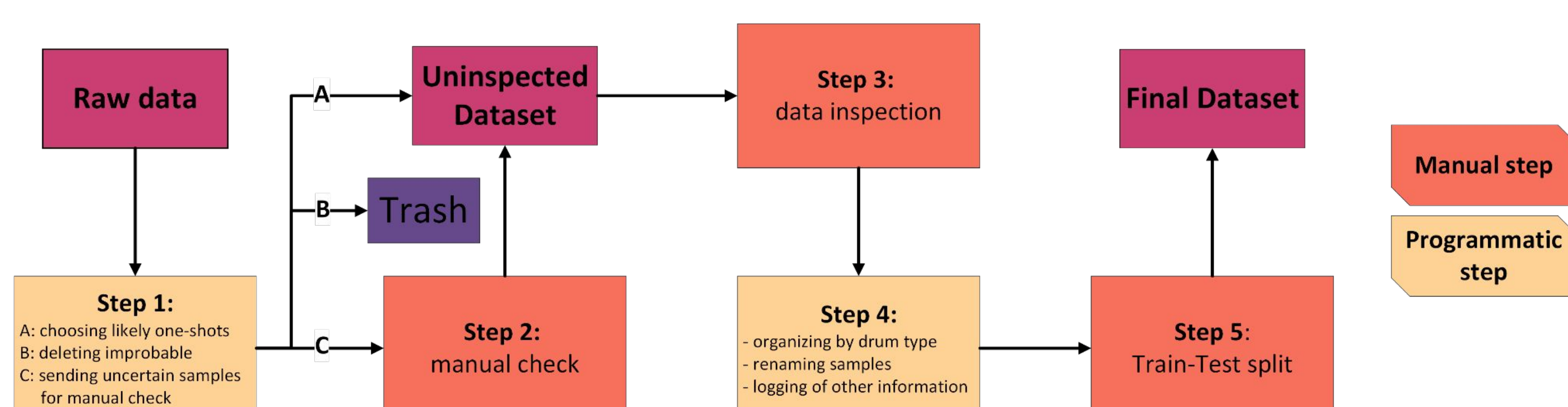


Electronic Music Production

- Nowadays, electronic music is produced using computer programs
- Various **one-shot samples** (short sounds of musical instruments, vocals, machine sounds ...) are organized into patterns, forming **loops** and **melodies** that are then connected into **track** (song)
- Meetings with artists clarified us the process of production
- Artists search for **one-shots** from extensive **libraries** containing thousands of samples, and it is a long and boring process

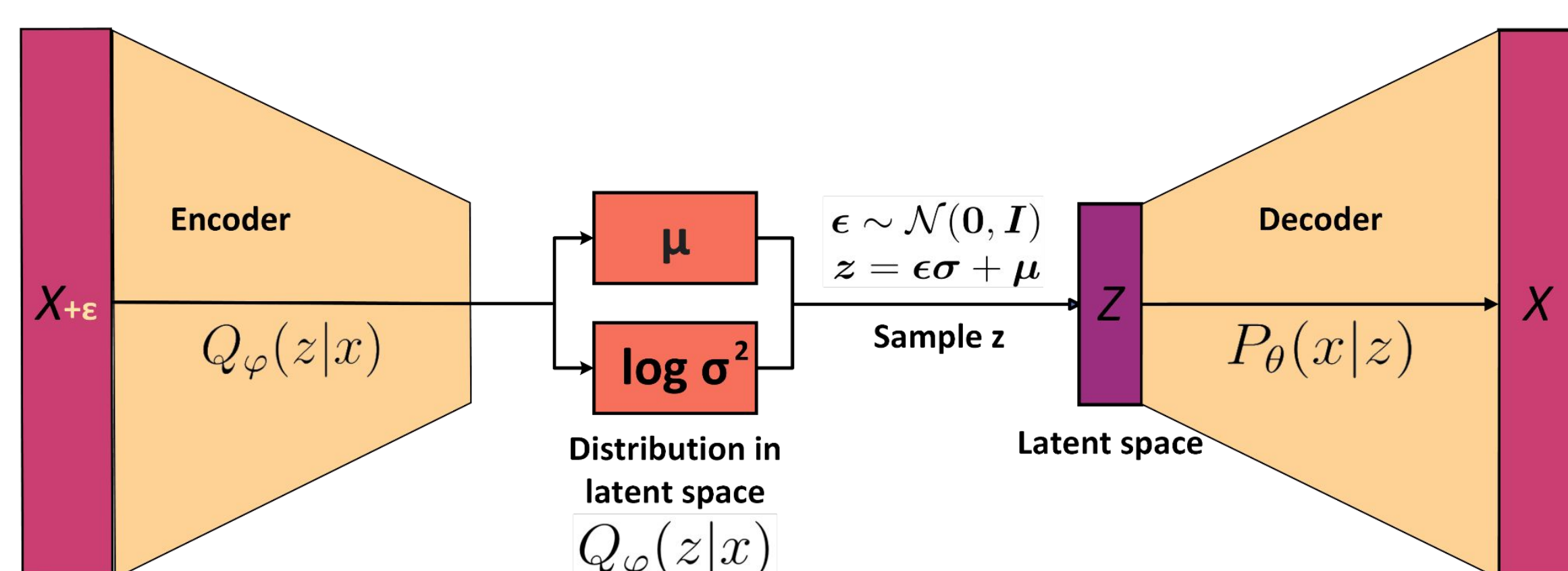
Dataset

- Almost **16 000 drum one-shot** samples organised in two directions:
 - **9 drum types** (part of drum kit the sounds originates from)
 - 11 electronic genre categories
- Dataset creation: download and **clean**



Model: Variational Autoencoder

- **Continuous coverage** of latent space
- Loss function has two parts that “fight” each other:
 - **reconstruction loss** (MSE)
 - **latent loss** implemented as **KL divergence**
 - **low** KL weight → poor generation
 - **high** KL weight → poor reconstruction

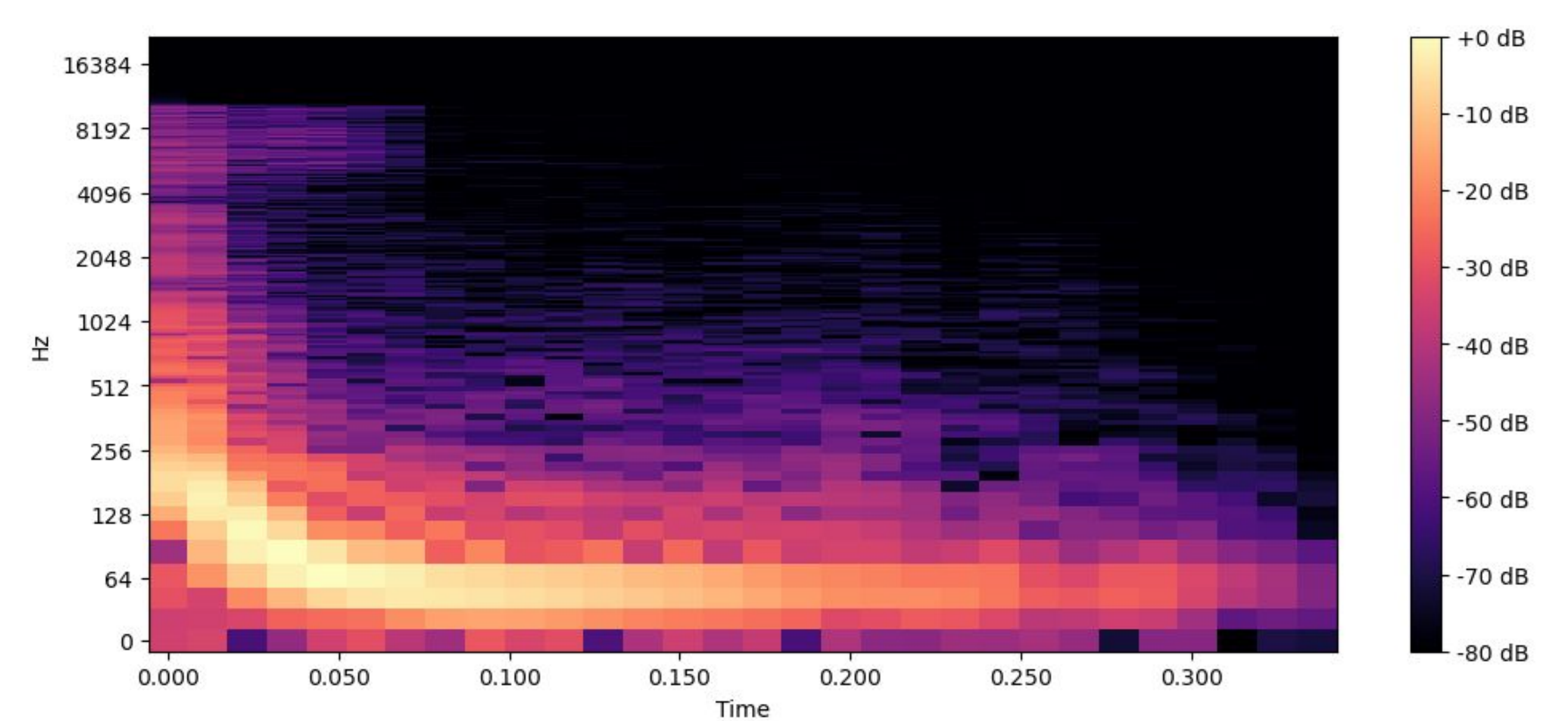


Our Goal is to create interactive generator of drum one-shot samples with machine learning methods

Sound representations: Spectrograms

- 2D time-frequency audio representations: **process sound as image**
 - Short-time Fourier transform - **STFT**
 - Mel-frequency cepstrum coefficients - **MFCCs**
- For generation, **invertibility** of representation is crucial
- Our representations are partially invertible with Griffin-lim alg.

STFT spectrogram of kick one-shot



Sample generation methods

- **Unconditionally sample latent space**, decode new sound
- **PCA shift**: move latent points by principal components
- **Reconstruction** is imperfect: can add interesting artefacts
- **Convex combination** (“kick + crash”): averaging points in latent space to find new sounds

Results

- Our best models (magpie) reconstruct decently
- Unconditional sampling sometimes produces interesting samples, but also it produces ugly noise
- First components of PCA **control high-level properties** such as **pitch, volume, decay length**.
- Convex combinations **create new sounds** that still relate to the original ones
- Next steps: better phase estimation, VST plugin

LISTEN HERE

